Lattice Boltzmann Formulation of the Demagnetising-Field Problem for a Non-Collinear (MRSh) Ferrofluid

MRT collision, single sub-step, $\nabla^2\psi = \nabla\cdot M$

Derivation, boundary conditions, and error analysis

Abstract

The magnetostatic closure $\nabla\cdot(\mu\nabla\psi) = 0$ used in phase-field lattice Boltzmann ferrofluid models presupposes $\boldsymbol{M} \parallel \boldsymbol{H}$ and is therefore inadmissible once the Martsenyuk–Raikher–Shliomis (MRSh) magnetisation dynamics is retained, since $\boldsymbol{M}\times\boldsymbol{H} \neq 0$ is the entire content of the rotational-viscosity effect. This document derives, from first principles, a multiple-relaxation-time lattice Boltzmann scheme for the resulting Poisson problem $\nabla^2\psi = \nabla\cdot\boldsymbol{M}$. The three free constants of the scheme — the lattice diffusivity D, the source weight c_Φ, and the macroscopic shift a — are determined rather than assumed; the wall condition is derived from a discrete flux balance rather than calibrated; the discrete solvability condition is obtained as an exact identity; and the temporal response of the pseudo-time solver is characterised so that a single sub-step per fluid step is provably sufficient. All error estimates are closed-form; numerical values are quoted only as sanity checks.

I. THE CONTINUUM PROBLEM

A. Magnetostatics

For a non-conducting magnetic fluid with no free currents, Maxwell's equations reduce to

$$\nabla\times\boldsymbol{H} = 0, \qquad \nabla\cdot\boldsymbol{B} = 0, \tag{1}$$

which state that $\boldsymbol{H}$ is irrotational and $\boldsymbol{B}$ solenoidal. In SI units the constitutive relation is

$$\boldsymbol{B} = \mu_0\,(\boldsymbol{H}+\boldsymbol{M})\,. \tag{2}$$

Irrotationality admits a scalar potential. We split the field into an externally imposed part $\boldsymbol{H}_0(t)$, which is uniform and hence divergence-free, and an induced (demagnetising) part:

$$\boldsymbol{H} = \boldsymbol{H}_0(t) - \nabla\psi, \tag{3}$$

so that ψ carries *only* the field generated by the magnetised fluid itself. Substituting (2) and (3) into $\nabla\cdot\boldsymbol{B} = 0$ and using $\nabla\cdot\boldsymbol{H}_0 = 0$ gives the governing elliptic equation

$$\boxed{\nabla^2\psi = \nabla\cdot\boldsymbol{M}} \tag{4}$$

This is a constant-coefficient Poisson equation: the permeability has disappeared, and with it every spatially varying coefficient.

B. Why the susceptibility closure is inadmissible

The model equation of the reference phase-field work,

$$\nabla\cdot(\mu\nabla\psi) = 0, \qquad \mu = \mu_0(1+\chi), \tag{5}$$

is algebraically equivalent to (4) *if and only if* $\boldsymbol{M} = \chi \boldsymbol{H}$ with scalar χ, i.e. $\boldsymbol{M} \parallel \boldsymbol{H}$. The MRSh closure retained here,

$$\boldsymbol{M} = \phi\, \varphi_p M_s\, L(\xi_e)\, \frac{\boldsymbol{\xi}_e}{\xi_e}, \tag{6}$$

with $\boldsymbol{\xi}_e$ the effective field obeying its own relaxation dynamics, is non-collinear by construction: $\boldsymbol{M} \times \boldsymbol{H} \neq 0$ is precisely the torque that produces $\Delta\eta$. Hence (5) contradicts the magnetisation model and must be replaced by (4). Here ϕ is the phase-field order parameter and φ_p the particle volume fraction.

C. Source term in the two-phase case

Writing $\boldsymbol{M} = \phi\, \boldsymbol{M}_f$ with $\boldsymbol{M}_f$ the magnetisation of the pure ferrofluid phase,

$$\nabla \cdot \boldsymbol{M} = \phi\, \nabla \cdot \boldsymbol{M}_f + \boldsymbol{M}_f \cdot \nabla\phi. \tag{7}$$

The second term is the interfacial bound magnetic charge. It vanishes identically in the present single-phase channel ($\phi \equiv 1$) but dominates once droplets are introduced, and is the reason the source must be evaluated with the same isotropic stencil already used for $\nabla\phi$.

D. Boundary conditions and solvability

On the boundary of the magnetic domain, magnetic insulation means that no flux of the *induced* field crosses, $\hat{\boldsymbol{n}} \cdot (\boldsymbol{H} - \boldsymbol{H}_0 + \boldsymbol{M}) = 0$, which with (3) reads

$$\left.\frac{\partial \psi}{\partial n}\right|_{\partial\Omega} = \gamma, \qquad \gamma \equiv \hat{\boldsymbol{n}} \cdot \boldsymbol{M}, \tag{8}$$

$\hat{\boldsymbol{n}}$ being the outward unit normal. We retain γ symbolically throughout, so that the alternative choice $\gamma = 0$ is recovered as a special case. Integrating (4) over Ω and applying the divergence theorem gives the solvability (compatibility) condition of the pure Neumann problem,

$$\oint_{\partial\Omega} \gamma \,\mathrm{d}A = \int_{\Omega} \nabla \cdot \boldsymbol{M} \,\mathrm{d}V = \oint_{\partial\Omega} \hat{\boldsymbol{n}} \cdot \boldsymbol{M} \,\mathrm{d}A. \tag{9}$$

With $\gamma = \hat{\boldsymbol{n}} \cdot \boldsymbol{M}$ this is an identity and the problem is always solvable. With $\gamma = 0$ it demands $\oint \hat{\boldsymbol{n}} \cdot \boldsymbol{M} = 0$, which is violated whenever the magnetisation is misaligned at the walls; the problem then has *no* solution and ψ grows without bound. The solution of (4) is unique only up to an additive constant, which is immaterial because only $\nabla\psi$ enters (3).

II. PSEUDO-TIME EMBEDDING AND LATTICE PRELIMINARIES

A. Embedding

A lattice Boltzmann equation cannot integrate an elliptic operator directly. Equation (4) is therefore embedded as the steady state of a diffusion equation in a fictitious time t:

$$\frac{\partial \psi}{\partial t} = D\nabla^2\psi + S, \qquad S \equiv -D\nabla \cdot \boldsymbol{M}. \tag{10}$$

Setting $\partial_t \psi = 0$ returns (4) *for any* $D > 0$, because the same D appears in both terms. This is the load-bearing property of the scheme: if the D multiplied into S differs from the D the lattice actually realises, the steady state becomes $\nabla^2\psi = (D_{\mathrm{code}}/D_{\mathrm{lattice}})\nabla \cdot \boldsymbol{M}$, a silent multiplicative error on every demagnetising field. Determining D_{lattice} exactly for the MRT operator is therefore the first task of the derivation.

B. Velocity set and moment identities

The D2Q9 set is used, with $c = \Delta x/\Delta t$ and $c_s^2 = c^2/3$:

$$\boldsymbol{e}_k = \begin{cases} (0,0), & k=0, \\ c\left(\cos\frac{(k-1)\pi}{2},\ \sin\frac{(k-1)\pi}{2}\right), & k=1,\dots,4, \\ \sqrt{2}\,c\left(\cos\frac{(2k-1)\pi}{4},\ \sin\frac{(2k-1)\pi}{4}\right), & k=5,\dots,8, \end{cases} \qquad w_k = \begin{cases} 4/9, \\ 1/9, \\ 1/36. \end{cases} \tag{11}$$

Four weight identities are used below; they follow from the antipodal symmetry of the set:

$$\sum_k w_k = 1, \qquad \sum_k w_k e_{k\alpha} = 0, \qquad \sum_k w_k e_{k\alpha} e_{k\beta} = c_s^2 \delta_{\alpha\beta}, \qquad \sum_k w_k e_{k\alpha} e_{k\beta} e_{k\gamma} = 0. \tag{12}$$

C. Equilibrium

The potential ψ is not advected, so the equilibrium carries no velocity coupling:

$$h_k^{\mathrm{eq}} = w_k\, \psi. \tag{13}$$

Its moments follow directly from (12):

$$\sum_k h_k^{\mathrm{eq}} = \psi, \qquad \sum_k e_{k\alpha} h_k^{\mathrm{eq}} = 0, \tag{14}$$

$$\sum_k e_{k\alpha} e_{k\beta} h_k^{\mathrm{eq}} = c_s^2 \psi\, \delta_{\alpha\beta}, \qquad \sum_k e_{k\alpha} e_{k\beta} e_{k\gamma} h_k^{\mathrm{eq}} = 0. \tag{15}$$

The vanishing first moment is what removes advection; the second moment is what becomes the Laplacian.

D. Source and macroscopic definition, with undetermined constants

The source is distributed on the weights and the macroscopic potential is allowed a shift, both with constants to be *derived*:

$$\Phi_k = c_\Phi\, w_k\, S, \qquad \psi \equiv \sum_k h_k + a\, S. \tag{16}$$

Their moments are

$$\sum_k \Phi_k = c_\Phi S, \qquad \sum_k e_{k\alpha} \Phi_k = 0. \tag{17}$$

No assumption $c_\Phi = 1$, $a = 0$ is made at this stage.

E. MRT evolution

With M the orthogonal moment matrix of the reference model and $\mathsf{S}_h = \mathrm{diag}(s_0, \dots, s_8)$,

$$\Lambda = \mathsf{M}^{-1} \mathsf{S}_h \mathsf{M}, \qquad \mathsf{S}_h = \mathrm{diag}\left(s_0, s_1', s_2', s_1, s_1, s_1, s_1, s_7', s_8'\right), \tag{18}$$

where $s_1 \equiv s_3 = s_5$ is the first-moment relaxation rate. The evolution equation is

$$h_k(\boldsymbol{x} + \boldsymbol{e}_k \Delta t, t + \Delta t) - h_k(\boldsymbol{x}, t) = -\Lambda_{k\ell}\left[h_\ell - h_\ell^{\mathrm{eq}}\right](\boldsymbol{x}, t) + \Delta t\, \Phi_k(\boldsymbol{x}, t). \tag{19}$$

III. THE MRT PROJECTION LEMMA

The Chapman–Enskog analysis of (19) requires only two contractions of Λ. Both follow from the structure of M: its row 0 is $M_{0k} = 1$ for all k, its row 3 is $M_{3k} = e_{kx}/c$, and its row 5 is $M_{5k} = e_{ky}/c$. Using $\sum_k M_{ak} (\mathsf{M}^{-1})_{kb} = \delta_{ab}$,

$$\sum_k \Lambda_{k\ell} = \sum_a s_a M_{a\ell} \sum_k (\mathsf{M}^{-1})_{ka} = \sum_a s_a M_{a\ell}\, \delta_{0a} = s_0, \tag{20}$$

$$\sum_k e_{k\alpha}\Lambda_{k\ell} = \sum_a s_a M_{a\ell} \sum_k M_{(\alpha)k}(\mathsf{M}^{-1})_{ka}\, c = s_1\, e_{\ell\alpha}, \tag{21}$$

with $(\alpha) = 3$ for $\alpha = x$ and $(\alpha) = 5$ for $\alpha = y$, and $s_3 = s_5 \equiv s_1$. In words: *the zeroth moment relaxes at s_0 and the first moment at s_1; no other relaxation rate enters the recovered macroscopic equation.* The rates s_2', s_7', s_8' act only on ghost modes and are free. This lemma is what makes the MRT case tractable in the same algebra as BGK, to which it reduces for $s_0 = s_1 = 1/\tau$.

IV. CHAPMAN–ENSKOG ANALYSIS

A. Taylor expansion

Define the link derivative $\mathscr{D}_k \equiv \partial_t + \boldsymbol{e}_k \cdot \nabla$. Expanding the streaming operator and dividing by Δt,

$$\mathscr{D}_k h_k + \frac{\Delta t}{2}\mathscr{D}_k^2 h_k + O(\Delta t^2) = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{\mathrm{neq}} + \Phi_k, \qquad h_k^{\mathrm{neq}} \equiv h_k - h_k^{\mathrm{eq}}. \tag{22}$$

The second-order term $\frac{\Delta t}{2}\mathscr{D}_k^2 h_k$ is retained; it is the origin of the $-\Delta t/2$ in the diffusivity and cannot be discarded.

B. Expansion and ordering

With δ the Knudsen bookkeeping parameter,

$$h_k = h_k^{(0)} + \delta h_k^{(1)} + \delta^2 h_k^{(2)}, \qquad \partial_t = \delta\partial_{t_1} + \delta^2\partial_{t_2}, \qquad \nabla = \delta\nabla_1, \qquad \Phi_k = \delta\Phi_k^{(1)}, \quad S = \delta S_1. \tag{23}$$

The source is ordered at $O(\delta)$; this is the general case and is what determines a and c_Φ uniquely. Collecting orders in (22):

$$O(\delta^0): \qquad h_k^{(0)} = h_k^{\mathrm{eq}} = w_k \psi, \tag{24}$$

$$O(\delta^1): \qquad \Delta t\, \mathscr{D}_k^{(1)} h_k^{(0)} = -\Lambda_{k\ell} h_\ell^{(1)} + \Delta t\, \Phi_k^{(1)}, \tag{25}$$

$$O(\delta^2): \qquad \partial_{t_2} h_k^{(0)} + \mathscr{D}_k^{(1)} h_k^{(1)} + \frac{\Delta t}{2}\big(\mathscr{D}_k^{(1)}\big)^2 h_k^{(0)} = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{(2)}, \tag{26}$$

with $\mathscr{D}_k^{(1)} = \partial_{t_1} + \boldsymbol{e}_k \cdot \nabla_1$.

C. Elimination of the second derivative

Applying $\mathscr{D}_k^{(1)}$ to (25) and substituting into (26) removes the $(\mathscr{D}_k^{(1)})^2$ term:

$$\partial_{t_2} h_k^{(0)} + \mathscr{D}_k^{(1)} h_k^{(1)} - \frac{1}{2}\mathscr{D}_k^{(1)}\big[\Lambda_{k\ell} h_\ell^{(1)}\big] + \frac{\Delta t}{2}\mathscr{D}_k^{(1)}\Phi_k^{(1)} = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{(2)}. \tag{27}$$

D. Constraints from the macroscopic definition

Expanding $\psi = \sum_k h_k + aS$ order by order, and using $\sum_k h_k^{(0)} = \psi$,

$$\sum_k h_k^{(1)} = -a S_1, \qquad \sum_k h_k^{(2)} = 0. \tag{28}$$

A non-zero a therefore makes the first-order populations carry non-zero mass; this is the non-standard ingredient of the scheme.

E. Moment balances

First moment of (25), using (14), (15), (17) and lemma (21), defines the diffusive flux $J_\alpha \equiv \sum_k e_{k\alpha} h_k^{(1)}$:

$$\Delta t\, c_s^2\, \partial_{1\alpha}\psi = -s_1 J_\alpha \qquad \Longrightarrow \qquad J_\alpha = -\frac{\Delta t\, c_s^2}{s_1}\, \partial_{1\alpha}\psi. \tag{29}$$

Zeroth moment of (25), using (28) and lemma (20):

$$\partial_{t_1}\psi = \left(\frac{s_0 a}{\Delta t} + c_\Phi\right) S_1. \tag{30}$$

At the fast scale the potential responds only to the source; diffusion is a t_2 effect.
Zeroth moment of (27), using both lemmas, (28) and (29):

$$\partial_{t_2}\psi + \left(1 - \frac{s_1}{2}\right)\partial_{1\alpha} J_\alpha + \left[\frac{s_0 a}{2} - a + \frac{\Delta t}{2} c_\Phi\right]\partial_{t_1} S_1 = 0, \tag{31}$$

and since $\left(1-\frac{s_1}{2}\right)/s_1 = \frac{1}{s_1} - \frac{1}{2}$,

$$\partial_{t_2}\psi = \Delta t\, c_s^2 \left(\frac{1}{s_1} - \frac{1}{2}\right)\nabla_1^2\psi + \left[a\left(1 - \frac{s_0}{2}\right) - \frac{\Delta t}{2} c_\Phi\right]\partial_{t_1} S_1. \tag{32}$$

F. Recovered equation and closure of the constants

Recombining $\partial_t = \delta\partial_{t_1} + \delta^2\partial_{t_2}$ from (30) and (32):

$$\boxed{\partial_t\psi = \underbrace{\Delta t\, c_s^2\left(\frac{1}{s_1} - \frac{1}{2}\right)}_{D}\nabla^2\psi + \underbrace{\left(\frac{s_0 a}{\Delta t} + c_\Phi\right)}_{\mathscr{C}} S + \underbrace{\left[a\left(1 - \frac{s_0}{2}\right) - \frac{\Delta t}{2} c_\Phi\right]}_{\mathscr{T}}\partial_t S} \tag{33}$$

Matching to the target (10) yields three statements:

$$\text{(C1) diffusivity:} \qquad D = \Delta t\, c_s^2\left(\frac{1}{s_1} - \frac{1}{2}\right) = \Delta t\, c_s^2\left(\tau_h - \frac{1}{2}\right), \quad \tau_h \equiv 1/s_1, \tag{34}$$

$$\text{(C2) source normalisation:} \qquad \mathscr{C} = \frac{s_0 a}{\Delta t} + c_\Phi = 1, \tag{35}$$

$$\text{(C3) no spurious transient:} \qquad \mathscr{T} = a\left(1 - \frac{s_0}{2}\right) - \frac{\Delta t}{2} c_\Phi = 0. \tag{36}$$

Three consequences deserve emphasis.

(i) D depends on s_1 *only*. The zeroth-moment rate s_0 does not enter it. In particular the familiar BGK formula $D = c_s^2(\tau - \frac{1}{2})\Delta t$ remains correct in MRT provided τ is read as $1/s_3 = 1/s_5$.

(ii) (C2) and (C3) together are solved uniquely. Substituting $c_\Phi = 1 - s_0 a/\Delta t$ into (C3), the terms $-s_0 a/2$ cancel on both sides and

$$a = \frac{\Delta t}{2}, \qquad c_\Phi = 1 - \frac{s_0}{2}. \tag{37}$$

(iii) (C2) alone is what fixes the steady state, because $\partial_t S \to 0$ there. Hence a second admissible closure exists,

$$a = 0, \qquad c_\Phi = 1 \qquad \Longrightarrow \qquad \mathscr{C} = 1, \quad \mathscr{T} = -\frac{\Delta t}{2}, \tag{38}$$

which recovers the exact steady state but retains a residual $-\frac{\Delta t}{2}\partial_t S$. Section 8 shows that (38) is the more accurate of the two whenever the drive is slow compared with the lattice relaxation, which is the regime enforced here. Any other pair violates (C2) and rescales every demagnetising field by $\mathscr{C} \neq 1$.

G. Recovery of the magnetic field

Since $\sum_k e_k h_k^{\mathrm{eq}} = 0$, the first moment of h is purely non-equilibrium, and (29) inverts to

$$-\partial_\alpha \psi = \frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{k\alpha} h_k \qquad \Longrightarrow \qquad H_\alpha = H_{0\alpha}(t) + \frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{k\alpha} h_k. \tag{39}$$

The first moment of (27) gives the leading correction to (39),

$$\sum_k e_{k\alpha} h_k^{(2)} = \tau_h^2 c_s^2 \left(1 - \frac{s_1}{2}\right) \partial_{1\alpha}(\partial_{t_1} \psi) \;\propto\; \partial_\alpha S, \tag{40}$$

which vanishes identically at convergence, $\partial_t \psi \to 0$. Equation (39) is therefore exact in the quasi-static limit and, unlike a finite-difference gradient, requires no ghost nodes at the walls.

V. BOUNDARY CONDITIONS

A. Specular reflection is a zero-flux condition

Let a flat wall lie halfway between the boundary node $\boldsymbol{x}_b$ and its image, with outward unit normal $\hat{\boldsymbol{n}}$. Define the specular map σ and the incoming set:

$$\boldsymbol{e}_{\sigma(k)} = \boldsymbol{e}_k - 2(\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}}, \qquad w_{\sigma(k)} = w_k, \qquad K_{\mathrm{in}} = \{k : \; \boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} < 0\}. \tag{41}$$

Pure specular reflection, $h_k(\boldsymbol{x}_b, t+\Delta t) = h^*_{\sigma(k)}(\boldsymbol{x}_b, t)$ for $k \in K_{\mathrm{in}}$, transfers

$$\sum_{k \in K_{\mathrm{in}}} h_k(\boldsymbol{x}_b, t+\Delta t) - \sum_{k \in K_{\mathrm{out}}} h_k^*(\boldsymbol{x}_b, t) = 0, \tag{42}$$

because σ is a bijection $K_{\mathrm{in}} \leftrightarrow K_{\mathrm{out}}$ preserving the weights. Specular reflection therefore imposes exactly $\partial\psi/\partial n = 0$, *not* magnetic insulation.

B. Derivation of the flux condition

Augment the reflection with an additive term proportional to the weights,

$$h_k(\boldsymbol{x}_b, t+\Delta t) = h^*_{\sigma(k)}(\boldsymbol{x}_b, t) + B\, w_k, \qquad k \in K_{\mathrm{in}}. \tag{43}$$

By (42) the net potential-mass entering per step is now

$$\Delta\Psi_{\mathrm{wall}} = B\, W_{\mathrm{in}}, \qquad W_{\mathrm{in}} \equiv \sum_{k \in K_{\mathrm{in}}} w_k. \tag{44}$$

The macroscopic flux of ψ is Fickian, $\boldsymbol{J} = -D\nabla\psi$, so the mass entering the domain through unit wall area per unit time is $-\hat{\boldsymbol{n}} \cdot \boldsymbol{J} = D\,\partial\psi/\partial n = D\gamma$. Equating,

$$\boxed{B = \frac{\Delta t\, D\, \gamma}{W_{\mathrm{in}}} = \frac{\Delta t\, D\, (\hat{\boldsymbol{n}} \cdot \boldsymbol{M})}{W_{\mathrm{in}}}} \tag{45}$$

For a flat D2Q9 wall three links enter, $W_{\mathrm{in}} = \frac{1}{9} + \frac{1}{36} + \frac{1}{36} = \frac{1}{6}$, so

$$B = 6\,\Delta t\, D\, (\hat{\boldsymbol{n}} \cdot \boldsymbol{M})\big|_{\mathrm{wall}}. \tag{46}$$

For the channel, $\hat{\boldsymbol{n}} = -\hat{\boldsymbol{y}}$ at the bottom and $\hat{\boldsymbol{n}} = +\hat{\boldsymbol{y}}$ at the top:

$$B_{\mathrm{bot}} = -6\,\Delta t\, D\, M_y^{w,\mathrm{bot}}, \qquad B_{\mathrm{top}} = +6\,\Delta t\, D\, M_y^{w,\mathrm{top}}. \tag{47}$$

Because the wall sits at $y = \mp\frac{1}{2}$ rather than on a node, γ must be extrapolated to the wall to preserve second-order accuracy:

$$M_y^{w,\mathrm{bot}} = \tfrac{3}{2} M_y\big|_{j=0} - \tfrac{1}{2} M_y\big|_{j=1}, \qquad M_y^{w,\mathrm{top}} = \tfrac{3}{2} M_y\big|_{j=n_Y-1} - \tfrac{1}{2} M_y\big|_{j=n_Y-2}. \tag{48}$$

Setting $\gamma = 0$ in (45) recovers plain specular reflection, confirming consistency.

C. Interior and curved boundaries

Equation (45) is stated for a general $\hat{\boldsymbol{n}}$ and is therefore directly reusable at obstacle surfaces: only W_{in} and $\hat{\boldsymbol{n}}$ change. For a staircase obstacle the flat-wall value $W_{\text{in}} = 1/6$ applies on faces and $W_{\text{in}} = \frac{1}{9} + \frac{1}{36}\cdot 3$ type sums apply at corners; these must be evaluated per node from the actual incoming set rather than assumed. Curved boundaries additionally require the wall value of $\hat{\boldsymbol{n}} \cdot \boldsymbol{M}$ at the true intersection point, i.e. an interpolation weight replacing the fixed $\frac{3}{2}, -\frac{1}{2}$ of (48).

VI. DISCRETE SOLVABILITY

Let $\Psi = \sum_{\boldsymbol{x}} \sum_k h_k$ be the total lattice mass. Summing the collision step over k and using lemma (20) together with $\sum_k h_k - \psi = -aS$ from (16),

$$\sum_k h_k^* = \sum_k h_k + \big(s_0 a + \Delta t\, c_\Phi\big) S = \sum_k h_k + \Delta t\, \mathscr{C}\, S = \sum_k h_k + \Delta t\, S, \tag{49}$$

the last step by (C2). Streaming conserves Ψ except at the walls, where (44) and (45) give $\Delta t\, D\gamma$ per boundary node. A steady state therefore exists if and only if

$$0 = \Delta t \sum_{\boldsymbol{x}} S + \Delta t\, D \sum_{\text{walls}} \gamma \qquad \Longleftrightarrow \qquad \boxed{\sum_{\boldsymbol{x}} \big(\nabla \cdot \boldsymbol{M}\big) = \sum_{\text{walls}} \gamma} \tag{50}$$

which is the discrete image of (9). Note that D cancels and c_Φ does not appear: the condition is a property of the discretisation alone.

Since the isotropic nine-point stencil used for $\nabla \cdot \boldsymbol{M}$ and the extrapolation (48) are different approximations, (50) is satisfied only to truncation order. The residual is removed by projecting the source onto the range of the discrete Laplacian:

$$\Delta \equiv \sum_{\boldsymbol{x}} \big(\nabla \cdot \boldsymbol{M}\big) - \sum_{\text{walls}} \gamma, \qquad \big(\nabla \cdot \boldsymbol{M}\big) \longleftarrow \big(\nabla \cdot \boldsymbol{M}\big) - \frac{\Delta}{N}, \tag{51}$$

N being the number of fluid nodes. This is the standard treatment of the singular pure-Neumann pressure-Poisson system in projection methods, where it is preferred to pinning a Dirichlet value at one node. The quantity Δ/N must be monitored: it is a truncation diagnostic when $\gamma = \hat{\boldsymbol{n}} \cdot \boldsymbol{M}$, but becomes an $O(1)$ fictitious magnetic charge when γ is set to zero in violation of (9).

VII. TEMPORAL RESPONSE AND THE SINGLE-SUB-STEP CONDITION

Linearising (10) for a Fourier mode of wavenumber k driven at angular frequency ω gives the exact transfer function

$$\hat{\psi} = \frac{\hat{S}}{\mathrm{i}\omega + Dk^2} = \frac{\hat{\psi}_{\text{qs}}}{1 + \mathrm{i}\,\omega\,\tau_{\text{relax}}}, \qquad \hat{\psi}_{\text{qs}} = \frac{\hat{S}}{Dk^2}, \tag{52}$$

where $\hat{\psi}_{\text{qs}}$ is the quasi-static (true magnetostatic) response. The lattice relaxation time is defined spectrally,

$$\tau_{\text{relax}} \equiv -\frac{\Delta t}{\ln |\lambda_2|}, \tag{53}$$

λ_2 being the second-largest eigenvalue modulus of the one-step operator of (19). It possesses two asymptotic branches,

$$\tau_{\text{relax}} \sim \frac{1}{D k_{\text{min}}^2} \ \text{(hydrodynamic, small } \varepsilon), \qquad \tau_{\text{relax}} \sim \tau_h = \frac{\varepsilon}{c_s^2} + \frac{1}{2} \ \text{(kinetic, large } \varepsilon), \tag{54}$$

writing $D = \varepsilon$ as in the implementation. The two branches cross near $\varepsilon \simeq c_s/k_{\text{min}}$, so τ_{relax} possesses a *minimum* in ε: increasing the relaxation parameter beyond that point makes the solver slower, not faster. There is consequently a floor on τ_{relax} that no choice of ε can defeat.

A single sub-step per fluid step is therefore sufficient if and only if the drive is slow compared with that floor:

$$\boxed{\omega\,\tau_{\text{relax}} \ll 1 \qquad \Longleftrightarrow \qquad \tau_B \gg (\omega\tau_B)\,\tau_{\text{relax}}} \tag{55}$$

since $\omega = (\omega\tau_B)/\tau_B$ and τ_B expressed in lattice steps is a free nondimensionalisation choice. Condition (55) is not a numerical convenience: it is the statement that the numerical magnetostatic relaxation must be fast compared with every physical time scale, which is the faithful lattice representation of instantaneous magnetostatics.

VIII. ERROR ANALYSIS

Four error sources are separated. Each is given in closed form; the numerical column is a sanity check only, obtained from exact steady-state linear solves and eigenvalue computations of the scheme, not from the production code.

A. E_1: spatial truncation

The Chapman–Enskog truncation at $O(\delta^3)$ leaves a fourth-derivative remainder, so the steady solution is second-order accurate,

$$E_1 = \frac{\|\psi_h - \psi\|_\infty}{\|\psi\|_\infty} = C_1\,(k\Delta x)^2 + O\big((k\Delta x)^4\big). \tag{56}$$

The constant depends on the closure. Since the shift contributes $a|S| = \frac{\Delta t}{2} D k^2 |\psi|$ directly to the reported potential,

$$C_1\big(a = \tfrac{\Delta t}{2}\big) - C_1\big(a = 0\big) = \frac{D\,\Delta t}{2}. \tag{57}$$

B. E_2: temporal lag

From (52), exactly,

$$E_2^{\text{amp}} = 1 - \frac{1}{\sqrt{1 + (\omega\tau_{\text{relax}})^2}} \simeq \frac{(\omega\tau_{\text{relax}})^2}{2}, \qquad E_2^{\text{phase}} = \arctan\big(\omega\,\tau_{\text{relax}}\big). \tag{58}$$

The closure (38) carries an additional transient error from $\mathscr{T} = -\frac{\Delta t}{2}$, of relative size $\omega\Delta t/2$.

C. E_3: source normalisation

Violating (C2) rescales the steady state exactly:

$$\nabla^2\psi = \mathscr{C}\,\nabla\cdot\boldsymbol{M}, \qquad E_3 = |\mathscr{C} - 1| = \left|\frac{s_0 a}{\Delta t} + c_\Phi - 1\right|. \tag{59}$$

D. E_4: compatibility defect

The projection (51) shifts the source uniformly by Δ/N, i.e. adds a fictitious uniform magnetic charge of that density:

$$E_4 = \frac{|\Delta|/N}{|\nabla\cdot\boldsymbol{M}|}. \tag{60}$$

E. Choice between the two admissible closures

Combining (57) with the transient term of (38), the two closures trade

$$E\big(a = \tfrac{\Delta t}{2}\big) - E\big(a = 0\big) \;\simeq\; \frac{\Delta t}{2}\Big(D k^2 - \omega\Big), \tag{61}$$

so that

$$\boxed{a=0,\ c_\Phi=1 \ \text{ is preferable whenever } \ \omega < Dk^2} \tag{62}$$

which is precisely the quasi-static condition (55) already imposed. The half-source shift is required only when the drive approaches the lattice relaxation rate; in the regime enforced here it adds more spatial error than the transient error it removes.

F. Numerical sanity check

Quantity	Prediction	Measured	Test
D vs. (34)	$\Delta t\, c_s^2(1/s_1-\frac{1}{2})$, indep. of s_0	ratio 0.998–1.0001 over $s_1\in[0.08,1.5]$	D-free source
Order of E_1	2	2.00, 2.00, 2.00 ($n_Y=16\to128$)	manufactured ψ
C_1 gap, (57)	$D\Delta t/2=2.5$ at $\varepsilon=5$	$3.771-1.045=2.73$	manufactured ψ
E_3, closure (37)/(38)	0	≤ 0.036 (= E_1)	manufactured ψ
E_3, ($c_\Phi=1-\frac{s_0}{2}, a=0$)	0.500	0.495	manufactured ψ
E_3, ($c_\Phi=1, a=\frac{\Delta t}{2}$)	0.500	0.541	manufactured ψ
Wall term (46)	exact	rel. err 2.6×10^{-15}	exact quadratic
Solvability (50), $\gamma=\hat{\boldsymbol{n}}\cdot\boldsymbol{M}$	$E_4\to0$	3×10^{-22}	slab $M_y(y)$
Solvability (50), $\gamma=0$	$E_4=O(1)$	$E_4=1.00$	slab $M_y(y)$
τ_{relax} minimum, (54)	at $\varepsilon\simeq c_s/k_{\min}$	min at $\varepsilon=5$, $\tau_{\text{relax}}=30$	spectral radius

Two entries deserve comment. First, the $\gamma=0$ row: because $\nabla\cdot\boldsymbol{M}=\partial_y M_y$ is single-signed across a channel, imposing $\partial\psi/\partial n=0$ forces the projection (51) to remove the entire mean of the source, $E_4=1$. Second, the spectral radius of the one-step operator was found to be exactly unity in every case tested — the conserved mode — so the floor in τ_{relax} is an accuracy limit, never a stability limit.

IX. SUMMARY OF GOVERNING EQUATIONS

For reference, the complete set to be implemented, in the notation of the preceding sections.

$$\boldsymbol{H}=\boldsymbol{H}_0(t)-\nabla\psi, \qquad \text{field split, applied plus induced} \tag{63}$$

$$\nabla^2\psi=\nabla\cdot\boldsymbol{M}, \qquad \text{magnetostatic potential} \tag{64}$$

$$\nabla\cdot\boldsymbol{M}=\phi\,\nabla\cdot\boldsymbol{M}_f+\boldsymbol{M}_f\cdot\nabla\phi, \qquad \text{bulk plus interfacial charge} \tag{65}$$

$$\frac{\partial\psi}{\partial t}=D\nabla^2\psi+S, \qquad S=-D\nabla\cdot\boldsymbol{M}, \qquad \text{pseudo-time embedding} \tag{66}$$

$$h_k^{\text{eq}}=w_k\psi, \qquad \text{equilibrium} \tag{67}$$

$$\Phi_k=c_\Phi w_k S, \qquad \text{source distribution} \tag{68}$$

$$\psi=\sum_k h_k+aS, \qquad \text{macroscopic potential} \tag{69}$$

$$h_k(\boldsymbol{x}+\boldsymbol{e}_k\Delta t,t+\Delta t)=h_k-\left(\mathsf{M}^{-1}\mathsf{S}_h\mathsf{M}\right)_{k\ell}\left(h_\ell-h_\ell^{\text{eq}}\right)+\Delta t\,\Phi_k, \qquad \text{MRT evolution} \tag{70}$$

$$D=\Delta t\,c_s^2\left(\frac{1}{s_1}-\frac{1}{2}\right), \quad s_1=s_3=s_5, \qquad \text{lattice diffusivity} \tag{71}$$

$$\frac{s_0 a}{\Delta t}+c_\Phi=1, \qquad \text{source normalisation (mandatory)} \tag{72}$$

$$a\left(1-\frac{s_0}{2}\right)=\frac{\Delta t}{2}c_\Phi, \qquad \text{transient removal (optional)} \tag{73}$$

$$H_\alpha=H_{0\alpha}(t)+\frac{1}{\tau_h c_s^2\Delta t}\sum_k e_{k\alpha}h_k, \qquad \text{field recovery} \tag{74}$$

$$h_k=h^*_{\sigma(k)}+B\,w_k, \quad k\in K_{\text{in}}, \qquad \text{wall closure} \tag{75}$$

$$B=\frac{\Delta t\,D\,(\hat{\boldsymbol{n}}\cdot\boldsymbol{M})}{W_{\text{in}}}, \qquad \text{wall flux term} \tag{76}$$

$$\sum_{\boldsymbol{x}} (\nabla \cdot M) = \sum_{\text{walls}} \hat{\boldsymbol{n}} \cdot M, \qquad \text{discrete solvability} \tag{77}$$

$$\omega\, \tau_{\text{relax}} \ll 1, \qquad \text{single-sub-step validity} \tag{78}$$

X. IMPLEMENTABLE FORM

Lattice units throughout: $\Delta x = \Delta t = 1$, $c = 1$, $c_s^2 = 1/3$, $\mu_0 = 1$. Indices (i, j) with i periodic and $j = 0, \ldots, n_Y - 1$ between halfway walls.

A. Relaxation set

$$s_1 = \frac{1}{\varepsilon/c_s^2 + 1/2} \equiv S_h^{(3)}, \qquad \tau_h = \frac{1}{s_1} = \frac{\varepsilon}{c_s^2} + \frac{1}{2}, \tag{79}$$

$$\mathsf{S}_h = \operatorname{diag}\big(s_0, 1.1, 1.1, s_1, s_1, s_1, s_1, 1.2, 1.2\big), \qquad s_0 = 1.0, \tag{80}$$

$$D = c_s^2 \left(\tau_h - \tfrac{1}{2}\right) = \varepsilon. \tag{81}$$

No permeability appears in (79): ε is a pseudo-time relaxation rate of dimension $[\mathrm{L}^2\mathrm{T}^{-1}]$, and the factor μ_0 carried over from the susceptibility closure is dimensionally inconsistent with (4).

B. Closure constants

Adopt (38), justified by (62):

$$c_\Phi = 1, \qquad a = 0 \qquad \Longrightarrow \qquad \psi_{i,j} = \sum_k h_{k,i,j}, \qquad \Phi_k = w_k S_{i,j}. \tag{82}$$

C. Source, with projection

$$\gamma_i^{\text{bot}} = -\left(\tfrac{3}{2} M_{y,i,0} - \tfrac{1}{2} M_{y,i,1}\right), \qquad \gamma_i^{\text{top}} = +\left(\tfrac{3}{2} M_{y,i,n_Y-1} - \tfrac{1}{2} M_{y,i,n_Y-2}\right), \tag{83}$$

$$\Delta = \sum_{i,j} (\nabla \cdot M)_{i,j} - \sum_i \left(\gamma_i^{\text{top}} + \gamma_i^{\text{bot}}\right), \qquad S_{i,j} = -D\left[(\nabla \cdot M)_{i,j} - \frac{\Delta}{n_X n_Y}\right]. \tag{84}$$

Note that $\gamma \equiv \hat{\boldsymbol{n}} \cdot M$ so $\gamma^{\text{bot}} = -M_y^w$ and $\gamma^{\text{top}} = +M_y^w$; the sum in (84) then reduces to $\sum_i (M_y^{w,\text{top}} - M_y^{w,\text{bot}})$. The divergence uses the same isotropic nine-point stencil already employed for $\nabla\phi$.

D. Collision and streaming

$$h^{\text{eq}}_{k,i,j} = w_k\, \psi_{i,j}, \tag{85}$$

$$h^{*}_{k,i,j} = h_{k,i,j} - \big(\mathsf{M}^{-1}\mathsf{S}_h\mathsf{M}\big)_{k\ell}\big(h_{\ell,i,j} - h^{\text{eq}}_{\ell,i,j}\big) + w_k S_{i,j}, \tag{86}$$

$$h_{k,i,j}(t+1) = \begin{cases} h^{*}_{k,\,i-e_{kx},\,j-e_{ky}}, & 0 \le j - e_{ky} \le n_Y - 1, \\ h^{*}_{\sigma(k),\,i-e_{kx},\,j_r} + B_{i-e_{kx}}\, w_k, & \text{otherwise}, \end{cases} \tag{87}$$

with the mirror index and wall terms

$$j_r = \begin{cases} -(j - e_{ky}) - 1, & j - e_{ky} < 0, \\ 2n_Y - 1 - (j - e_{ky}), & j - e_{ky} \ge n_Y, \end{cases} \tag{88}$$

$$B_i^{\text{bot}} = 6D\,\gamma_i^{\text{bot}}, \qquad B_i^{\text{top}} = 6D\,\gamma_i^{\text{top}}. \tag{89}$$

The x index is taken modulo n_X. Setting $B \equiv 0$ reduces (87) to pure specular reflection, i.e. $\partial\psi/\partial n = 0$.

E. Field recovery

$$H_{x,i,j} = H_{0x}(t) + \frac{1}{\tau_h c_s^2} \sum_k e_{kx} h_{k,i,j}, \qquad H_{y,i,j} = H_{0y}(t) + \frac{1}{\tau_h c_s^2} \sum_k e_{ky} h_{k,i,j}. \tag{90}$$

F. Ordering within one time step

Because $\nabla \cdot \boldsymbol{M}$ and the projection Δ are non-local, the field, magnetisation and source passes must be separated; the h update itself remains inside the same time step and is executed once.

$$\underbrace{(90)}_{\boldsymbol{H} \text{ from } h} \longrightarrow \underbrace{\text{MRSh}}_{\boldsymbol{M} \text{ from } \boldsymbol{H},\ \boldsymbol{\Omega}} \longrightarrow \underbrace{(83)\text{–}(84)}_{S \text{ from } \boldsymbol{M}} \longrightarrow \underbrace{(82), (85)}_{\psi,\ h^{\text{eq}}} \longrightarrow \underbrace{(86), (87)}_{\text{collide, stream}} \tag{91}$$

The magnetic forces and the f and g lattices are unaffected and retain their present position in the loop.

G. Parameter selection

Two conditions constrain the numerical scales.

$$\varepsilon: \ \text{minimise } \tau_{\text{relax}}(\varepsilon) = \max\left(\frac{1}{\varepsilon k_{\min}^2},\ \frac{\varepsilon}{c_s^2} + \frac{1}{2}\right) \implies \varepsilon \simeq \frac{c_s}{k_{\min}}, \tag{92}$$

$$\tau_B: \ \tau_B \gg (\omega \tau_B)\, \tau_{\text{relax}} \quad \text{[quasi-static, Eq. (55)]}. \tag{93}$$

A third, independent, physical constraint fixes the same scale. Eliminating the particle volume via $\tau_B = 3\eta V / k_B T$ and $\xi_0 = \mu_0 m H_0 / k_B T$,

$$\frac{\varphi_p M_s}{H_0} = \frac{3\,\eta\,\varphi_p\,\xi_0}{\mu_0 H_0^2\,\tau_B}, \tag{94}$$

which must match the experimental value of the fluid being reproduced. Since (93) and (94) both scale as τ_B^{-1}, they are satisfied together. Holding $\Omega \tau_B$ fixed under a change of τ_B requires the driving body force to scale as τ_B^{-1}.

H. Diagnostics to output

$$\Psi = \sum_{i,j} \sum_k h_{k,i,j} \qquad \text{bounded} \Leftrightarrow (50) \text{ holds; linear growth} \Rightarrow \text{no solution} \tag{95}$$

$$E_4 = \frac{|\Delta| / (n_X n_Y)}{|\nabla \cdot \boldsymbol{M}|} \qquad \text{fictitious charge injected by the projection} \tag{96}$$

$$\omega\, \tau_{\text{relax}} \qquad \text{validity of the single sub-step} \tag{97}$$

XI. VERIFICATION PROTOCOL

Before coupling, with $\boldsymbol{M}$ frozen:

1. **Manufactured solution.** Choose $\psi = \sin(k_x x) \cos\left(k_y (y + \frac{1}{2})\right)$ with $k_y = m\pi / n_Y$ so that $\gamma = 0$ at both walls, set $\nabla \cdot \boldsymbol{M} = \nabla^2 \psi$, and confirm $E_1 \propto \Delta x^2$. This is the single test that simultaneously validates D, c_Φ, a and s_0, because the converged ψ must be *independent of* ε.
2. **Non-zero wall flux.** Repeat with $\gamma \neq 0$ to exercise (89); the exact solution of $\psi'' = \text{const}$ with prescribed end slopes is quadratic and must be reproduced to round-off.
3. **Uniformly magnetised slab.** $\boldsymbol{M} = M_0 \hat{\boldsymbol{y}}$ inside the channel gives the exact result $H_y = -M_0$ (transverse demagnetising factor unity in a slab) and H_x undisturbed. A returned value $\mathscr{C} M_0$ with $\mathscr{C} \neq 1$ localises the failure to (72).
4. **Solvability.** Verify Ψ bounded and E_4 at round-off with $\gamma = \hat{\boldsymbol{n}} \cdot \boldsymbol{M}$; contrast with $\gamma = 0$, where Ψ must grow linearly at the rate $\Delta t \sum_{\boldsymbol{x}} S$.